

Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(\overset{\text{Predicted}}{\uparrow} h_{\theta}(x)^{(i)} - \overset{\text{Target}}{\uparrow} y^{(i)} \right)^2$$

↪ Error

Mean Squared Error

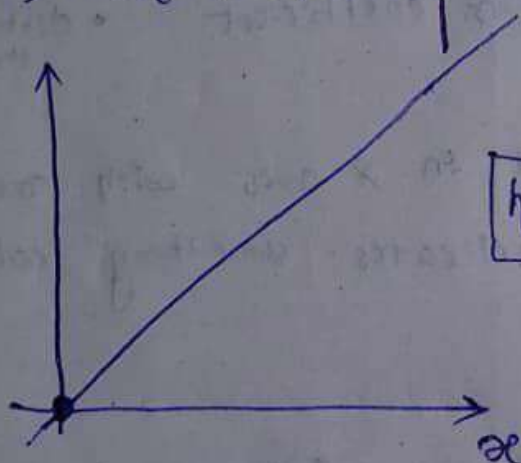
Final Aim, what we need to solve

Minimize $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x)^{(i)} - y^{(i)} \right)^2$

θ_0, θ_1

① $h_{\theta}(x) = \theta_0 + \theta_1 x$ ↪

$\theta_0 = 0$



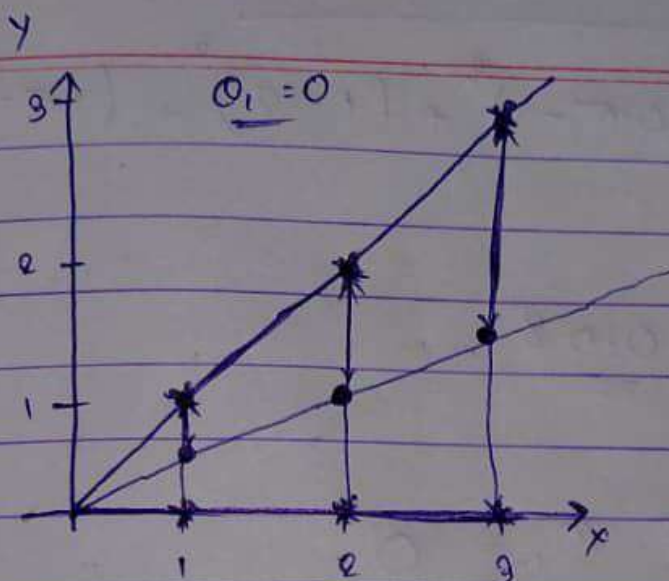
$h_0(x) = \theta_1 x$

DATASET

x	y
1	1
2	2
3	3

note: $m \rightarrow$ no. of points

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$$\theta_0 = 0$$

$$h_0(x) = \theta_0 x$$

let, $\theta_1 = 1$ & slope 1

$$h_0(x) = 1 \quad x=1$$

$$h_0(x) = 2 \quad x=2$$

$$h_0(x) = 3 \quad x=3$$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

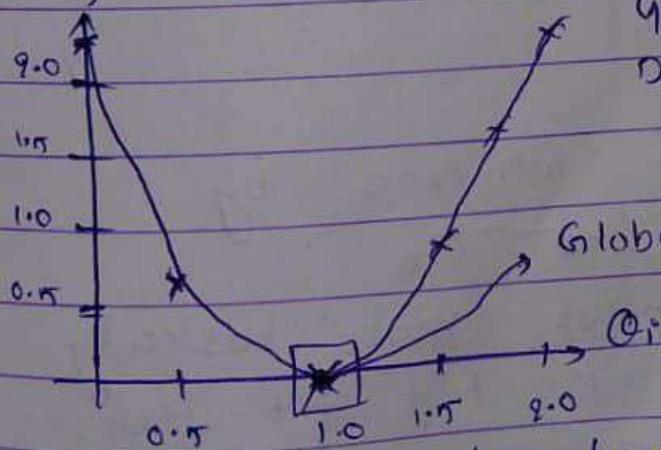
$$= \frac{1}{2 \times 3} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

if $\theta_1 = 1$

$$J(\theta_1) = 0$$

$$h_0(x) = \theta_1 x$$

$$J(\theta_1)$$



Gradient
Descent

let $\theta_1 = 0.5$

$$h_0(x) = 0.5 \quad \text{if } x=1$$

$$h_0(x) = 1 \quad \text{if } x=2$$

$$h_0(x) = 1.5 \quad \text{if } x=3$$

error has been minimised

$$\theta_1 \text{ or } \theta_0$$

$$\text{if } \theta_1 = 0.5$$

$$J(\theta_1) = \frac{1}{2 \times 3} \left[(0.5 - 1)^2 + (1 - 2)^2 + (1.5 - 3)^2 \right]$$

$$J(\theta_1) = \approx \underline{\underline{0.58}}$$

$$\text{if } \theta_1 = 0$$

$$J(\theta_1) = \frac{1}{2 \times 3} \left[(0 - 1)^2 + (0 - 2)^2 + (0 - 3)^2 \right]$$

$$J(\theta_1) = \approx 2.9$$

Note: Gradient Descent function is very important and its also used on DL.

We achieve global minima by changing the θ_1 or θ_0

If we get this point that basically indicate we achieve best fit line.